# Harik Sodhi

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## EDUCATION

**University of Oxford** | *MEng Engineering Science* **2025-2029**

- *Lab project:* Implemented RK4 integrator for rocket landing dynamics (thrust, drag, gravity); designed proportional controller and optimised 1000-dimensional velocity-height profile for fuel efficiency using constrained optimisation via penalty function (MATLAB Optimization Toolbox), with robustness testing across perturbed initial conditions

## EXPERIENCE

**Oxford Alpha Fund** **Oxford**
*Director* *January 2026 - Present*

- Selected as a Quant Analyst (top ˜10% of applicants) via competitive research project and presentation process
- Investigated volatility risk premium in short-dated BTC options; identified statistically significant day-of-week effects (Friday $p = 0.008$, Sunday $p < 0.001$) and constructed systematic straddle strategies achieving Sharpe 1.18 and 70% win rate `github.com/Harik-S/oaf-hilary-2026`
- Developed automated tooling to generate sponsor-ready CV books from member CVs, enabling rapid updates during recruitment cycles
- Evaluated applications and prepared candidates for internship interviews

**Optiver Trading Academy** **October 2026 - November 2026**
*Optiver* *Oxford*

- Selected for Optiver's competitive Trading Academy program as a first-year
- Traded ETFs and futures on Optiver's proprietary simulation platform, practising market making, delta hedging, and risk management under realistic market conditions, finishing with a positive P&L

## AWARDS & HONOURS

Silver Medal, International Physics Olympiad (49th/420 competitors from 89 countries; highest rank in Western Europe) 2025

UK International Physics Olympiad Team Member 2025

Purple Comet Mathematics Competition — UK National Champion 2024

Winner, Berkeley Math Tournament Online Power Round 2024

BMO1 Distinction 2024

## PROJECTS

### Market Making Simulator

- Modelled market making against optimal, suboptimal, and insider trader agents; analysed how player-type composition affects spreads and profitability

### DesmosArt

- Built a Python tool converting arbitrary drawings into parametric equations renderable on Desmos, using image processing and curve fitting